

► 1 Task-based approaches to automation in the literature

Central to the study of technology on work is the insight that jobs are a “bundle of tasks” (Autor, 2015). As such, task automation might, or might not lead to job automation, depending on the importance of a particular task to an occupation. Analyses of employment effects of automation that use the task-based approach, including this study, attempt to capture two central considerations: (i) the ability of a specific technology to automate a given task and (ii) the character of existing occupations in the labour market, based on the detailed tasks performed by these occupations. By modelling the intersection of these two dimensions, one arrives at a range of possible projections of how such technology could interact with occupational tasks, leading either to their full automation through technology-driven substitution, or to a transformation, through partial automation and human-machine complementarity. While the task-based framework certainly has limitations, recent academic literature shows the analytical advantages of this method for modelling employment and labour market impacts (Acemoglu et al., 2024; Nurski and Ruer, 2024).

Research on AI and tasks has shown important improvements in methods and scope in recent years, which includes using AI tools for research purposes. In one of the early papers on Machine Learning (ML) and jobs, Frey and Osborne, (2013) fed the opinions of a group of IT technology experts into an ML algorithm to develop occupational-level projections of automation potential for the US job market. The emergence of GenAI APIs has simplified the possibility of combining human expertise and algorithmic capacities for research purposes. Relying on this idea, Eloundou et al. (2023) demonstrated a close alignment of GPT-4 predictions with a survey of 70 AI experts on the potential of automating occupational tasks with LLMs, and subsequently built on this feature to develop synthetic automation scores for the US market.

Most task-based studies are focussed on the US, due to the very detailed nature of the O*NET database, as well as its public accessibility. This typically forces researchers interested in the global picture or other countries and regions to base their analysis on the strong assumption of similarity of local economic and labour effects to those projected in the US context. In reality, occupational tasks can vary significantly among individual countries, which poses a particular challenge in developing countries. As such, working with tasks in the ISCO-08 system offers a useful and simple common denominator for projections across different countries, regions and income groups. Our method combines the strengths of this global framework with a significantly more detailed national classification system in Poland, which provides a detailed list of occupations and tasks, based on the latest labour market data. This helps us establish a starting point for modelling interaction of GenAI’s abilities with up-to-date tasks and occupations in a setting located between the countries with highest incomes and the emerging economies (Figure 1). Using a combination of GenAI tools, we subsequently transfer this knowledge back to the tasks in ISCO-08 system.